

19.0 INSULATION

19.1 THERMAL INSULATION

- 19.1.1 Thermal insulation shall be applied to all piping, equipment, and ductwork detailed in this specification after completion and acceptance of all tests for leaks.
- 19.1.2 All thermal insulation shall be FM Global approved.
- 19.1.3 Insulation material shall be Closed Cell Elastomeric Nitrile Rubber/ Physically cross-linked polyolefin foam.
- 19.1.4 Insulation material shall be EPD Compliant and CFC HCFC Free. Insulation shall be Dust and fiber Free. Density of insulation material should be 40-60 kg /m³ Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.
- 19.1.5 Thermal conductivity of elastomeric nitrile rubber shall not exceed 0.037 W / m K @ 20 Deg C as per EN 12667/ EN ISO 8497
- 19.1.6 The insulation shall have fire performance such that it passes Class 1 as per BS476 Part 7 for surface spread of flame as per BS 476 and also pass Fire Propagation requirement as per BS476 Part 6 to meet the Class 'O' Fire category as per 1991 Building Regulations (England & Wales) and the Building Standards (Scotland) Regulations 1990.
- 19.1.7 Water vapor permeability shall not exceed 0.017 Perm inch (2.48 x 10⁻¹⁴ Kg/m.s.Pa), i.e. Moisture Diffusion Resistance Factor or 'μ' value should be minimum 7000 EN 12086 / EN 13469. Insulation shall not absorb water more than 0.2% by volume as per ASTM C 1763 / ASTM C 209.
- 19.1.8 The insulation shall be with self-adhesive for ducting and piping and available in roles / sheets.
- 19.1.9 All the refrigerant pipework and condensate pipework insulation shall be laminated with factory made glass cloth (Blue / Red Color/Grey/Black/Green) of 7 mil / 0.18 mm thickness along with manufacturer-recommended adhesive where the VOC level should not exceed more than 850 grams / Liter. Outdoor pipework installation glass cloth shall have UV resistant. The glass cloth shall have good puncture and tear resistance, shall have good UV resistance as per ASTM G 26A / ISO 4892-2 Method A.
- 19.1.10 All the indoor chilled water pipework insulation shall be laminated with factory made 7 mil / 0.18 mm glass cloth. Pipe work that is located exposed in abuse areas, within 8 ft. (2.5 m) of floor, or exterior to the building shall have an aluminum jacket. Pipe work outside of abuse areas and exposed to view shall have field applied PVC jackets color coded for system type. Pipe work outside of abuse areas and concealed shall have factory applied ASJ-SSL jackets with PVC fitting covers.

Complies



- 19.1.11 Thermal insulation on indoor conditioned/non-conditioned area supply/return ductwork, outdoor air duct where crossing conditioned area insulation shall be laminated with factory made 7 mil / 0.18 mm glass cloth.
- 19.1.12 Thermal insulation on outdoor directly exposed supply/return ductwork insulation shall be laminated with factory made 7 mil / 0.18 mm glass cloth, 26 G aluminium cladding with factory laminated moisture barrier (vapour retarder) for mechanical protection.
- 19.1.13 The thickness of the insulation shall be as specified for the individual application. Insulation shall be designed to avoid the condensation, heat loss. Contractor to carry out the insulation thickness calculation based on the selected material and shall be provided for the EoR approval. The estimated thickness of insulation as follows and shall be evaluated based on the final selection.
- 19.1.14 25 mm thick closed cell elastomeric nitrile rubber for Chilled Water Pipe.
- 19.1.15 9 mm thick closed cell elastomeric nitrile rubber for oval duct and shall be thermally pre-insulated from factory. This shall be insulated inside the ductwork.
- 19.1.16 13mm (indoor duct) and 25mm (outdoor ducts) thick closed cell elastomeric nitrile rubber for rectangular ducts shall be insulated from site.
- 19.1.17 The entire duct for air conditioning from AHU mouth shall be insulated with closed cell elastomeric nitrile rubber except the acoustically insulated duct pieces.
- 19.1.18 Outdoor pipes shall be insulated with 25mm for both gas and liquid pipes and 13mm (gas pipe)/ 19mm (liquid pipe) thick closed cell elastomeric nitrile rubber for refrigerant pipework and 9mm thick for condensate pipework.
- 19.1.19 All insulation on equipment and piping shall be applied only after the system has been pressure tested satisfactorily.
- 19.1.20 Duct surfaces shall be cleaned to remove all grease, oil, dirt, etc. prior to carrying out insulation work. Measurement of surface dimensions shall be taken properly to cut closed cell insulation to size with sufficient allowance in dimension. Cutting of insulation sheets shall be done with adjustable blade to make 900 cuts in thickness of sheet. Hacksaw or blades are not acceptable tools for cutting the insulation.
- 19.1.21 Where ducts/pipes penetrate walls / floor it shall be insulated with intumescent properties insulation material for fire protection. The treatment shall be minimum 500 mm extended on both sides.
- 19.1.22 The adhesives for the application of insulation should be of principal manufacturer. Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.

Complies



- 19.1.23 The EoR reserve the right to peruse the weights, dimensions etc., of the insulation material supplied. Samples of all insulation material specified, in various forms shall be submitted by the Contractor. The EoR and Client shall have the right to reject all supplies which do not conform to the specifications and to the samples so approved.

19.2 ACOUSTIC INSULATION

- 19.2.1 25mm thick open cell elastomeric nitrile rubber, class "1" secured with approved adhesive for acoustic insulation shall be used for ductwork (initial 5 meters on AHU Main supply duct) as indicated in HVAC layout. This shall be insulated inside the ductwork.
- 19.2.2 All acoustic insulation shall be FM Global approved.
- 19.2.3 The Random Incidence Sound Absorption Coefficient (RISAC) of the material - Tested as per ISO 354:2003, ISO 11654:1997 & ASTM C 423
- 19.2.4 The acoustic insulation material shall be dust and fiber free.
- 19.2.5 The acoustic insulation material shall have good sound absorption, damping and barrier properties.
- 19.2.6 The density of acoustic insulation material shall be within 140-180 Kg/m³.
- 19.2.7 The acoustic insulation material shall have a thermal conductivity not exceeding 0.047 W/(m.K) @ 20 Deg. C as per EN 12667.
- 19.2.8 The acoustic insulation material shall withstand maximum surface temperature of +85 Deg. C and minimum surface temperature of -20 Deg. C as per EN 14706.
- 19.2.9 The acoustic insulation material shall pass Air Erosion Resistance Test in accordance with ASTM Standard C 1071-05 without any laminate or covering (section 12.7), from an independent accredited testing agency.

Complies

19.3 UNDERDECK INSULATION

- 19.3.1 Thermal Insulation for Underdeck application using Class " 0 " Closed cell nitrile rubber elastomeric foam / Physically cross-linked polyolefin foam insulation laminated with chemically treated glass cloth (Grey Color) of 7 mil / 0.18 mm thickness along with manufacturer recommended adhesive Insulation material manufactured should have fire properties - Class 0 Rating as per BS 476 Part 7 & Part 6 test & HB and V0 Rating as per UL 94. Insulation shall have a minimum Thermal Conductivity of 0.035 W / m K @ 0 Deg C as per EN 12667/ EN ISO 8497.
- 19.3.2 Insulation shall have a Water Vapor Diffusion Resistance Factor of 7,000 as per EN 12086 / EN 13469. Insulation shall not absorb water more than 0.2% by volume as per ASTM C 1763 / ASTM C 209.

Complies



- 19.3.3 The insulation material shall be EPD Compliant and CFC HCFC Free. Insulation shall be Dust and Fiber Free. Adhesives should be used from the insulation manufacturer, Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.

19.4 OVERDECK INSULATION

- 19.4.1 Thermal Insulation for Overdeck application using FM approved Expanded polystyrene / polyisocyanurate foam insulation material shall be EPD Compliant and CFC HCFC Free.
- 19.4.2 Insulation shall be Dust and Fiber Free. Adhesives should be used from the insulation manufacturer, Application services to be availed from the principal insulation company and detailed report to be provided by the insulation manufacturer after installation.